



SIKORSKY UH-60M BLACKHAWK FOR X-PLANE



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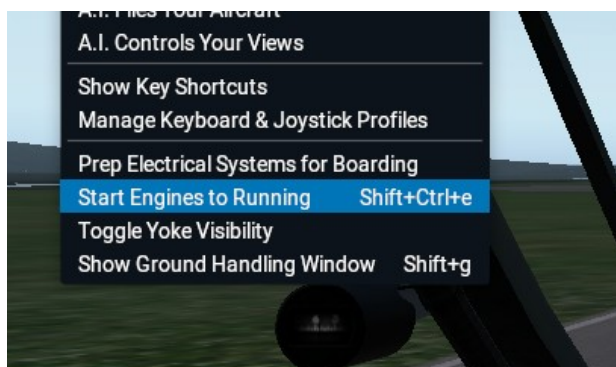
Missions

Datarefs and Commands



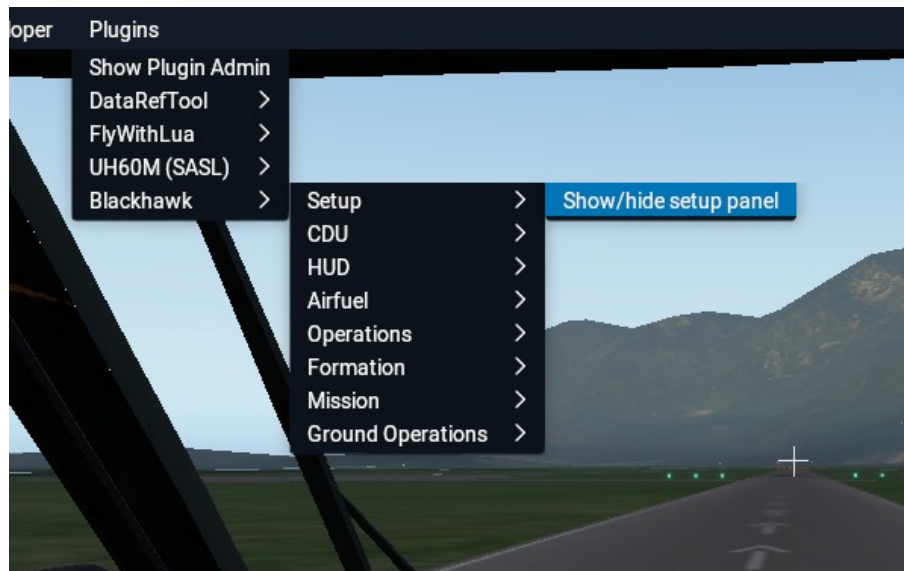
Quick Start

- Make sure you do not have your joystick mapped to a „throttle“ axis. If you have a hardware throttle, use the „wing sweep“ axis instead !
- Start the helicopter either with „running engines“ or use the „autostart“ option from the „Flight“ pull-down menu.



Configuration

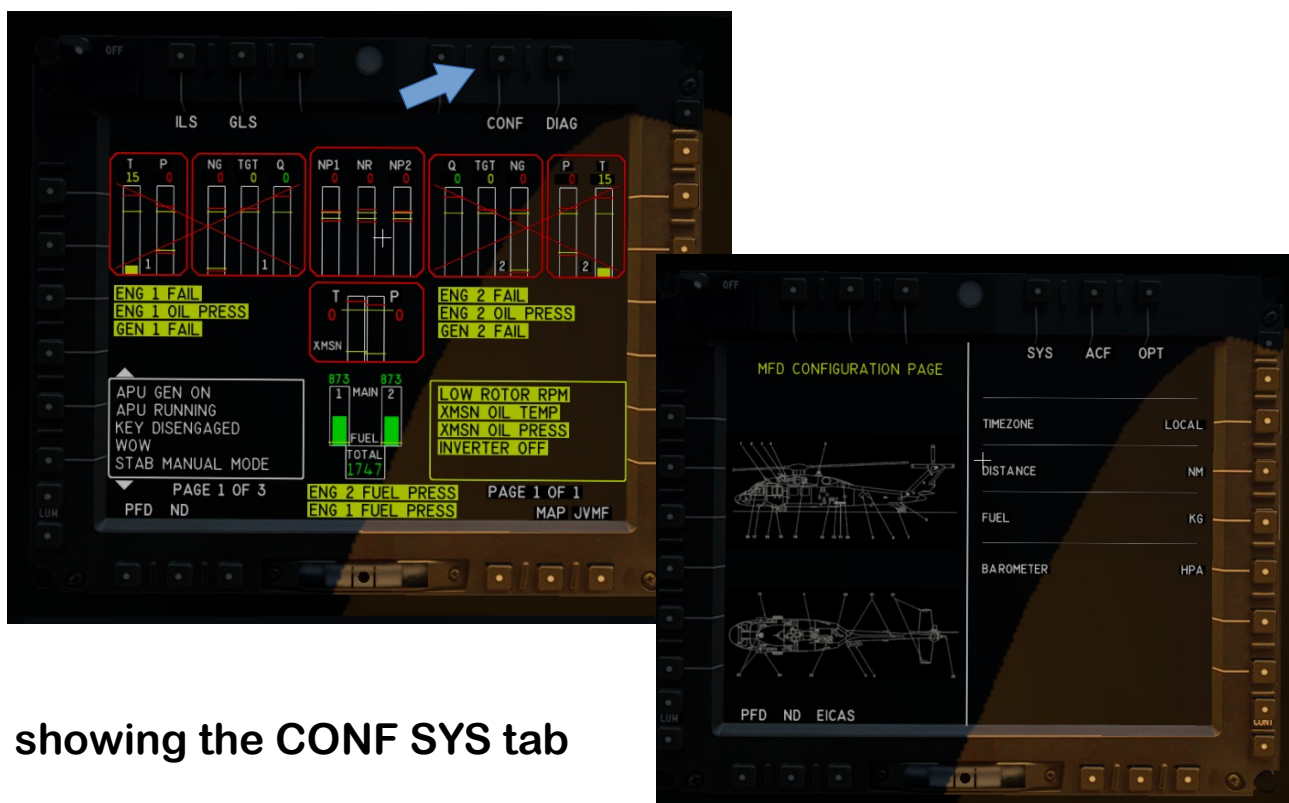
To configure the Blackhawk you can open the „setup panel“ from the „Blackhawk“ setup sub-menu inside the „Plugins“ pull-down menu.



Alternatively you can click on the upper-right-corner of the cockpit dash. If uncertain use the „Show instrument click regions“ under the „View“ menu.

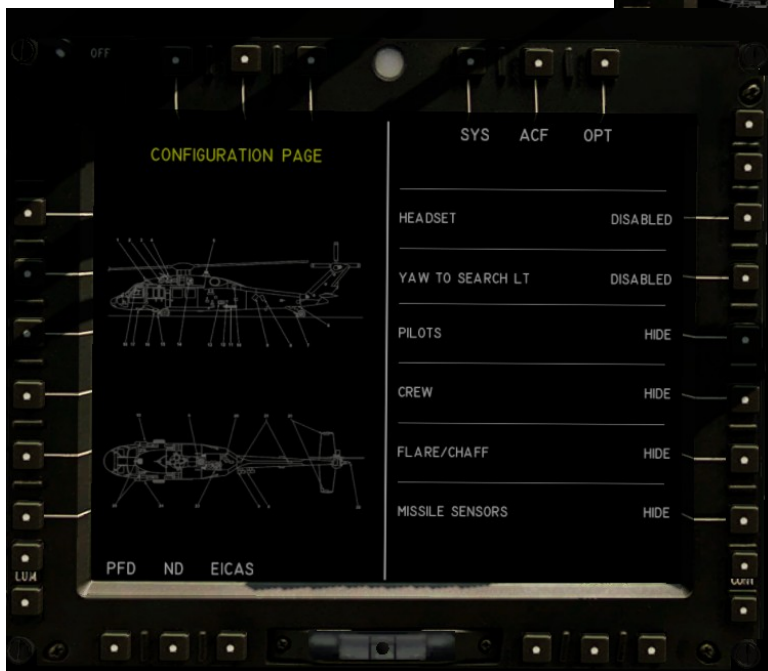
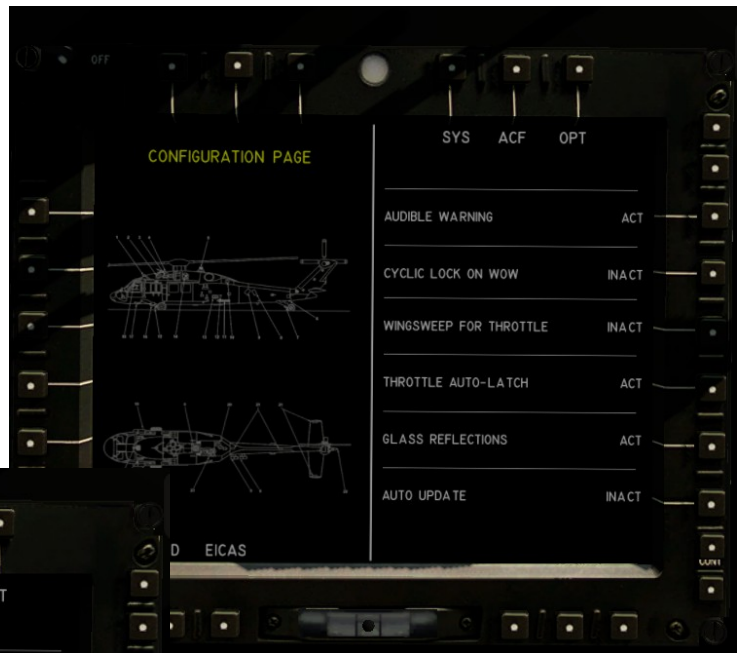


Other configurations can be done from the Multi-Function-Displays (MFD) under the „CONF“ button on the „EICAS“ screen (of course batteries have to be switched ON first).



showing the CONF SYS tab

the CONF ACF tab



and
the CONF OPT tab

settings and configurations are saved in either the acf_config.ini located in the acf folder or in the opt_config.ini located in the selected livery folder, depending whether they are „global“ or „livery“ related.

Cold & Dark

On the SETUP panel you can select the CHECKLIST option. A popup will guide you interactively through the steps required to start the helicopter manually from cold&dark.



At the end of the list the engine-start can either be performed by pressing the white buttons on the throttle levers located on the over-head panel, or by assigning the appropriate command to a joystick button (see “Datarefs and Commands” section).



Please note that the “Ignition Key” needs to be set to the “ON” position !



Check messages on the EICAS screen and confirm them by pressing the CAUTION button or via the mapped command. Audible alarm can be configured in the CONF menu on the MFD.



CDU – Control and Display Unit

FMS – Flight Management System

The CDU or FMS is being used to control communication and navigation systems, flight plans and missions. A mission is similar to a flight plan but has additional functionality such as running X-Plane commands depending on programmable conditions. More about it in the “Missions” section.

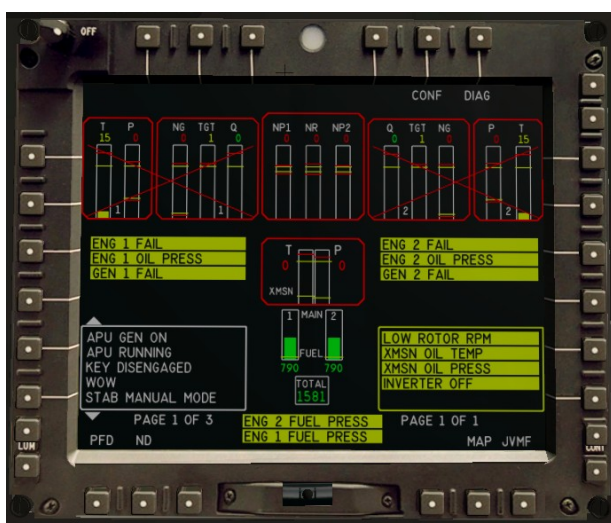
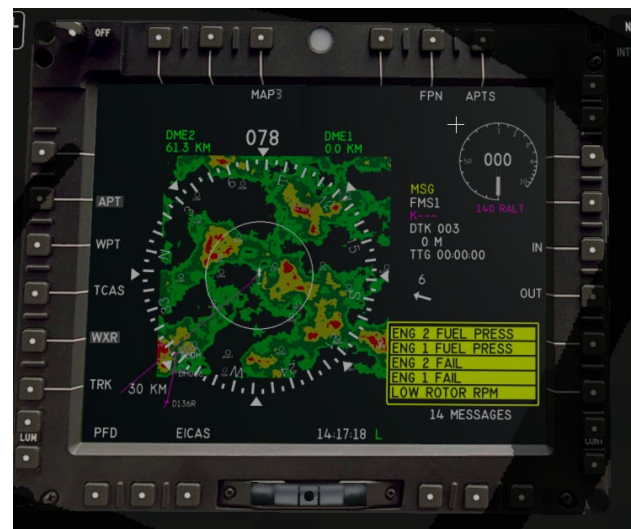
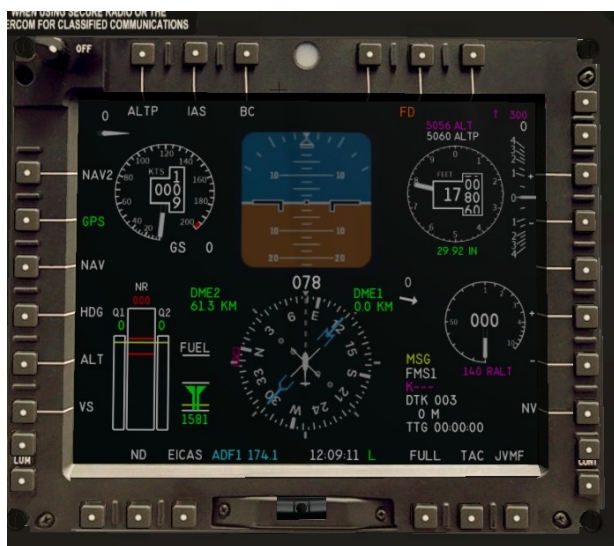


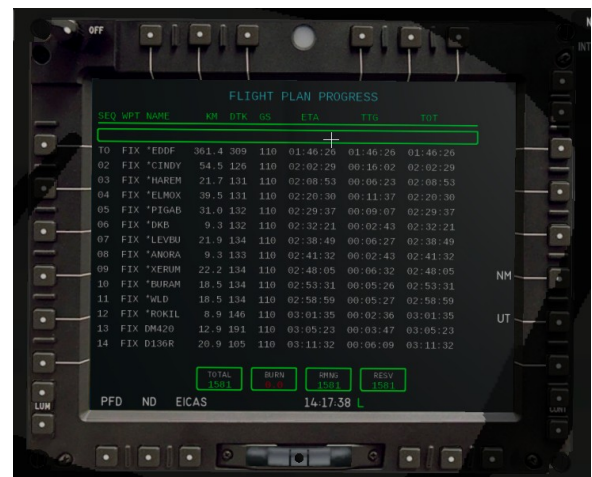
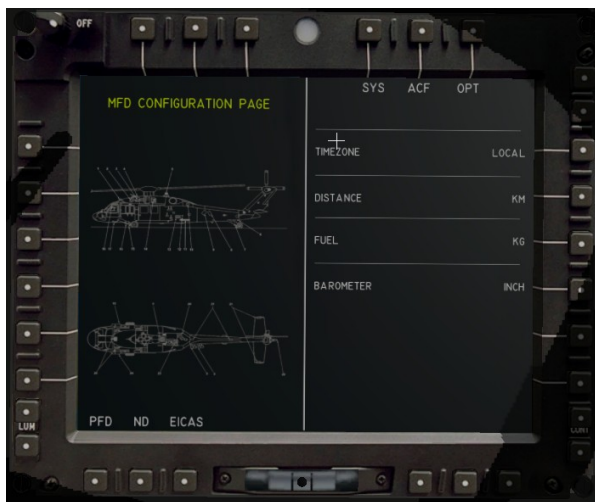


Clicking on the CDU displays in the pedestal will open a CDU popup. This can be helpful in VR. The popups are movable and resizable.



MFD – Multi Function Displays





Avitab support (also on iPad)



Camera guided target locking



JVMF MESSAGES

JVMF stands for Joined Variable Message Format.

It is kind of a military email system. In the simulation it is used to create "tasks" for the crew. Via the "Operations" pull down menu a task to extract a Seal from a random location can be created.

An OSM query in the background tries to find an appropriate spot in the area 20km around the current position. Once the message is being accepted (ACK = acknowledged), the FMS is being programmed with the coordinates and so can lead the pilot to the Seal.

The JVMF-softkey switches to **YELLOW** as soon as a new "unread" message shows up in the Inbox. The symbols ">" greater-than" and "?" indicate "un-read" and "un-acknowledged" messages.



See: https://www.youtube.com/watch?v=2KXv8P6lc_w

FDCP – Flight Director Control Panel



The FDCP is the control panel for the autopilot. The 5 rotaries for RALT (radio altimeter) , ALTP (programmed altitude). ALT (altitude), IAS (indicated airspeed) and HDG (heading) can be used to change the values in the displays. Pushing the rotaries syncs the display with the actual values (i.e. current altitude).

The black buttons under the displays allow to activate/deactivate the related “upper” AP (autopilot) function. If a function is activated it will be indicated by a green LED inside the button. Some function also have an “armed” state, which is indicated by an amber LED.

The very right button in the upper row controls and show whether the FD (flight director) is engaged/coupled. If it is coupled (CPLD) then the FD controls the aircraft.

The very left button in the upper row engages RNAV (radio navigation) which means the selected navigation source is providing input to the AP. Valid sources are NAV1, NAV2, LOC1, LOC2 , FMS. They can be selected with the outer ring of the very left rotary in the upper row.

“GO ARND” = GoAround will set the AP to climb with 700 ft/sec and 70 kts.

“LVL HOLD” = LevelHold enables the PID controller which keeps the current altitude-above-ground. It is armed (amber LED) unless vertical speed is less 0.1 m/sec and kias $\leq$ 50 kts. Then it will change to green.

“HVR POS” = HoverHold enables the PID pid controller which keeps the current position if the helicopter hovers with less the 1.5 m/sec horizontal speed.

NOTE: All AP functions are DISENGAGED if we have WOW (weight-on-wheels). This also applies to all weapons !!

Cameras

The Blackhawk comes with a built-in multi-camera system. It allows you to do role-playing games.



The system tracks each individual cam position and also allows to save, load and recall position sequences (i.e. preflight check)

The cameras can be selected through the setup-menu or by using one of the available commands (see DataRefs and Commands).

For example in combination with LVL-HOLD and HVR-HOLD you can observe HRST operations as a Crew Chief while sitting in the Cargo Area.

Hint: make sure you engage your AP before you leave your pilots seat ;)

Operations

HRST Fast-Rope-Extraction

Requirement: "HRST" option enabled in setup menu.

Command: uh60m/ops/drop_dude

Every time the command gets called, a Seal will slide down on one of the ropes and if on the ground, moves to a random "defense position". This also works on moving objects like the aircraft carrier.

Air-Refuel

Requirement: "REFUEL PROBE" enabled in setup menu.

Command: uh60m/ops/c130_show.

First command will call the C130. It will adopt to your heading and speed.

Second command will end operation. C130 will climb and disappear.

Third command will delete the C130 immediately.

Target tracking

Requirement: "FLIR CAM" enabled in setup menu.

On MFD1 select TAC and then TGT. Clicking the mouse into MFD display will show a red aiming mark which allows to control the cam with the mouse. Pressing the LCK button (or use the equiv. command) will lock the camera to the aimed position. A yellow pole will show up at the locked location. A 2nd click will release the target and the camera moves back to the default position.

Extract Seal

An OSM query is being used to select a random location in the range of 20km to place a Seal at. A JVMF order is being created to get him home. If the message is ACKnowledged, the FMS is programmed with the location. Fly to the Seal and land next to him to take him on board.

Lights

Search light

Use command “uh60m/lights/srch_Lt_on” to switch search lite on and off. Other commands to steer the search light: ‘srch_Lt_up’, ‘srch_Lt_dn’, ‘srch_Lt_lft’ and ‘srch_Lt_rgt’. On ground (wow) the search light is coupled to the yaw axis.

Landing light

Use command “uh60m/lights/ldg_Lt_on” to switch landing lite on and off.



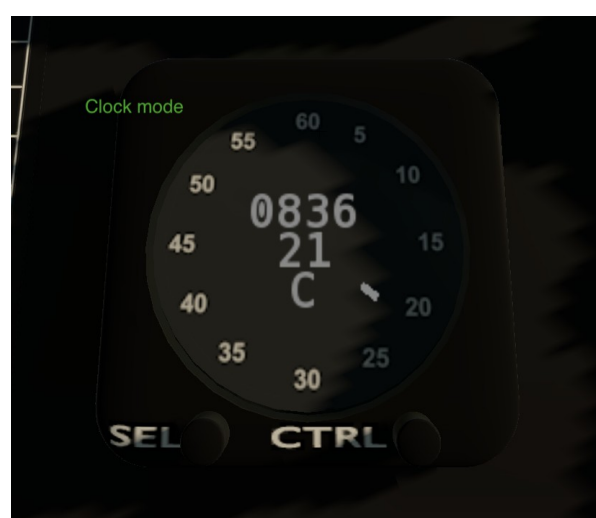
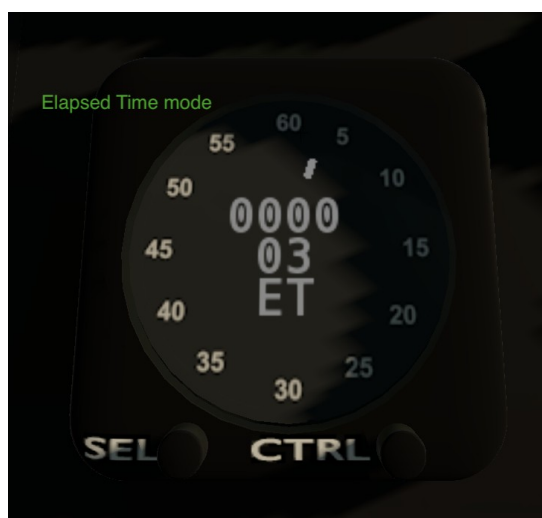
Instrument lights



Cockpit, Cabin dome and external lights



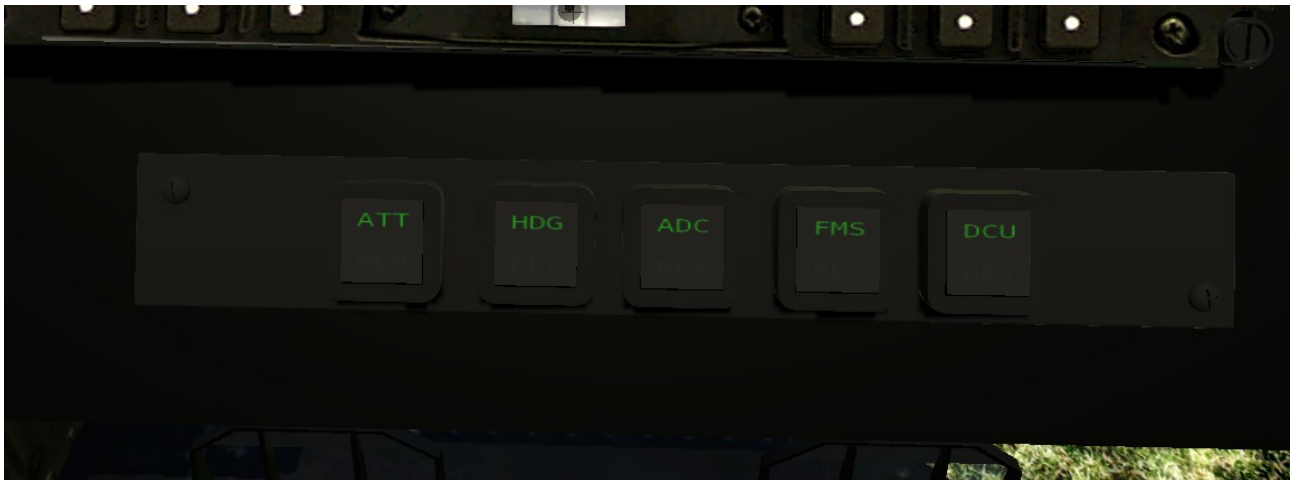
Clock



Reversionary switch panel

The UH-60M has a number of redundant systems. The reversionary switch panel allows the pilot and co-pilot to switch between the primary and the secondary system. In the simulation these functions are limited by the available options provided by X-Plane.

At the moment it is possible to switch the ADC (air data) only. Depending of the switch setting, the MFDs show altitude (baro and radio), air speed, vertical speed and barometer setting from the “pilot” (green) or the “co-pilot” (amber) instruments.



Weapons

GUNS

Requirement: “GUNS” enabled in setup menu

Commands: uh60m/weapons/arm_guns & uh60m/weapons/fire_guns

Both “gunner doors” need to be open !

The guns are tied to the yaw axis. So they will follow the tail rotor pedals.

CHAFF

Commands: uh60m/weapons/arm_flares & uh60m/weapons/deploy_chaff

FLARE

Commands: uh60m/weapons/arm_flares & uh60m/weapons/deploy_flares

Note: weapons are disabled as long as we have WOW (weight on wheels)

See also: <https://www.youtube.com/watch?v=zKQfuxRcJ8I>



GUNS



CHAFF/FLARES



EOMS



Videos

UH60M Videos playlist

<https://www.youtube.com/watch?v=MuvI8dFNZUo&list=PLtspyUICsgSZDCGbs1DjPPdm96kQ2mA7F>

HowTo-clips playlist:

<https://www.youtube.com/watch?v=AfriXZv96VE&list=PLtspyUICsgSaIVPmxsf2exIWePyU4dzOE>

Make sure you subscribe to this YouTube channel to not miss new videos

Missions

see MISSIONS_HOWTO.TXT in “missions” folder

Datarefs and Commands

Door commands

uh60m/doors/door_right_open	Cockpit doors
uh60m/doors/door_right_close	
uh60m/doors/door_right_toggle	
uh60m/doors/door_left_open	
uh60m/doors/door_left_close	
uh60m/doors/door_left_toggle	
uh60m/doors/cargo_right_open	Cargo doors
uh60m/doors/cargo_right_close	
uh60m/doors/cargo_right_toggle	
uh60m/doors/cargor_left_open	
uh60m/doors/cargo_left_close	
uh60m/doors/cargo_left_toggle	
uh60m/doors/gun_right_open	Gunner doors
uh60m/doors/gun_right_close	
uh60m/doors/gun_right_toggle	
uh60m/doors/gun_left_open	
uh60m/doors/gun_left_close	
uh60m/doors/gun_left_toggle	
uh60m/doors/avionics_open	Avionics bay door
uh60m/doors/avionics_close	
uh60m/doors/avionics_toggle	
uh60m/doors/hydraulics_open	Hydraulics bay door
uh60m/doors/hydraulics_close	
uh60m/doors/hydraulics_toggle	
uh60m/doors/engines_open	Engine cowlings
uh60m/doors/engines_close	
uh60m/doors/engines_toggle	
uh60m/doors/toggle_all	Toggle all doors
uh60m/doors/toggle_one	Select door by dataref
uh60m/doors/load_door_preset	dref: uh60m/doors/preset

Light commands

uh60m/lights/ldg_lt_on	Landing light toggle
uh60m/lights/srch_lt_on	Search light toggle
uh60m/lights/srch_lt_up	Search light up
uh60m/lights/srch_lt_dn	Search light down
uh60m/lights/srch_lt_lft	Search light left
uh60m/lights/srch_lt_rgt	Search light right
uh60m/lights/dome_blue	Cargo dome light blue
uh60m/lights/dome_white	Cargo dome light white
uh60m/lights/light_dn	Dim cockpit lights
uh60m/lights/light_up	Brighten cockpit lights
uh60m/lights/cockpit_flood	Bright flood light toggle

System and Engines commands

uh60m/engn/start_engine0	Start engine 1
uh60m/engn/start_engine1	Start engine 2
uh60m/engn/idle_stop0	Release idle stop latch 1
uh60m/engn/idle_stop1	Release idle stop latch 2
uh60m/sys/hover_hold	en/disable hover hold
uh60m/sys/level_hold	en/disable level hold
uh60m/sys/sas_boost	en/disable SAS boost
uh60m/sys/hstab_up	Move stabilator manually up
uh60m/sys/hstab_dn	Move stabilator manually down
uh60m/sys/clock_sel	Select clock/timer mode
uh60m/sys/clock_ctrl	start/stop/reset timer
uh60m/sys/ftr	Force Trim Release
uh60m/caution_ack	Acknowledge caution warnings
uh60m/menu	Open setup menu

OPS commands

uh60m/ops/winch_prep	
uh60m/ops/winch_ena	
uh60m/ops/winch_conn	
uh60m/ops/winch_up	
uh60m/ops/winch_down	
uh60m/ops/rescuer_hook_only	

uh60m/ops/ipad_home	Ipad home button
uh60m/ops/ipad_toggle	Ipad show toggle
uh60m/ops/fuel_probe_out	Extend fuel probe
uh60m/ops/fuel_probe_in	Retract fuel probe
uh60m/ops/fuel_probe_toggle	Toggle fuel probe
uh60m/ops/rope_out	Show ropes
uh60m/ops/rope_in	Hide ropes
uh60m/ops/drop_dude	Drop seal
uh60m/ops/extract_seal	Place Seal at OSM pos and set FMS
uh60m/ops/iff_emerg	Sets transponder to 7700

Nav commands

uh60m/nav/nav_src_up	Select next RNAV source
uh60m/nav/nav_src_dn	Select previous RNAV source

Weapon commands

uh60m/weapons/arm_guns	Arm GUNS
uh60m/weapons/fire_guns	Fire GUNS
uh60m/weapons/arm_flares	Arm CHAFF and FLARES
uh60m/weapons/deploy_flares	Deploy FLARES
uh60m/weapons/deploy_chaff	Deploy CHAFF

Camera commands

uh60m/cam/p1_select	Select Pilot camera
uh60m/cam/p2_select	Select CoPilot camera
uh60m/cam/c1_select	Select CrewChief camera
uh60m/cam/g1_select	Select ground crew 1 camera
uh60m/cam/g2_select	Select ground crew 2 camera
uh60m/cam/hoist_select	Select hoist camera
uh60m/cam/add	Add position to path
uh60m/cam/next	Go to next path position
uh60m/cam/reset	Go to first path position
uh60m/cam/clear	Clear path
uh60m/cam/save	Save paths to new.cam
uh60m/cam/update	Update current position
uh60m/cam/load	Load paths from default.cam
uh60m/cam/load_preflight	Load preflight path

MFD commands

uh60m/mfd/cam_power	FLIR Camera power
uh60m/mfd/cam_lock	FLIR Camera locked to target
uh60m/mfd/mfd1_pwr	Power button on MFD1
uh60m/mfd/mfd1_t1	MFD1 top row button 1
uh60m/mfd/mfd1_t2	MFD1 top row button 2 ... 6
uh60m/mfd/mfd1_r1	MFD1 right row button 1
uh60m/mfd/mfd1_r2	MFD1 right row button 2 6
uh60m/mfd/mfd1_b1	MFD1 bottom row button 1
uh60m/mfd/mfd1_b2	MFD1 bottom row button 2 ... 6
uh60m/mfd/mfd1_l1	MFD1 left row button 1
uh60m/mfd/mfd1_l2	MFD1 left row button 2 ... 6
uh60m/mfd/q1up	Queue 1 (yellow) msg box up
uh60m/mfd/q1dn	Queue 1 (yellow) msg box down
uh60m/mfd/q2up	Queue 2 (white) msg box up
uh60m/mfd/q2dn	Queue 2 (white) msg box down
uh60m/mfd/mfd_inc	Increment selected instrument
uh60m/mfd/mfd_dec	Decrement selected instrument
uh60m/mfd/mfd_inc2	Increment fast sel. Instr.
uh60m/mfd/mfd_dec2	Decrement fast sel. Instr.
uh60m/mfd/mfd_push	Push selected instrument
uh60m/mfd/mfd1_s1	MFD 1 select field 1
uh60m/mfd/mfd1_s2	MFD 1 select field 2
uh60m/mfd/mfd1_s3	MFD 1 select field 3
uh60m/mfd/mfd1_s4	MFD 1 select field 4
uh60m/mfd/mfd1_s5	MFD 1 select field 5
uh60m/mfd/mfd1_s6	MFD 1 select field 6

FDCP commands

uh60m/mfd/fdcp1d11	FDCP 1 Display 1 button 1
uh60m/mfd/fdcp1d12	FDCP 1 Display 1 button 2
uh60m/mfd/fdcp1d13	
uh60m/mfd/fdcp1d21	
uh60m/mfd/fdcp1d22	
uh60m/mfd/fdcp1d23	
uh60m/mfd/fdcp1d31	

uh60m/mfd/fdcp1d32	
uh60m/mfd/fdcp1d33	
uh60m/mfd/fdcp1d41	
uh60m/mfd/fdcp1d42	
uh60m/mfd/fdcp1d51	
uh60m/mfd/fdcp1d52	
uh60m/mfd/fdcp2d33	
uh60m/mfd/fdcp1_push1	Push rotary 1
uh60m/mfd/fdcp1_push2	Push rotary 2
uh60m/mfd/fdcp1_push3	Push rotary 3
uh60m/mfd/fdcp1_push4	Push rotary 4
uh60m/mfd/fdcp1_push5	Push rotary 5
uh60m/mfd/fdcp1_nxt_but	Select next button
uh60m/mfd/fdcp1_prv_but	Select previous button
uh60m/mfd/fdcp1_sel_but	Push selected button
uh60m/mfd/fdcp1_nxt_psh	Select next rotary
uh60m/mfd/fdcp1_prv_psh	Select previous rotary
uh60m/mfd/fdcp1_sel_psh	Push selected rotary
uh60m/mfd/fdcp1_rot_inc	Increment selected rotary
uh60m/mfd/fdcp1_rot_dec	Decrement selected rotary

CDU commands

uh60m/cdu/CDU1-popup	Open CDU1 popup
uh60m/cdu/CDU2-popup	Open CDU2 popup
uh60m/cdu/CDU1_L1	
uh60m/cdu/CDU1_L2	
uh60m/cdu/CDU1_L3	
uh60m/cdu/CDU1_L4	
uh60m/cdu/CDU1_L5	
uh60m/cdu/CDU1_R1	
uh60m/cdu/CDU1_R2	
uh60m/cdu/CDU1_R3	
uh60m/cdu/CDU1_R4	
uh60m/cdu/CDU1_R5	
uh60m/cdu/CDU1_CLR	

uh60m/cdu/CDU1_COM	
uh60m/cdu/CDU1_NAV	
uh60m/cdu/CDU1_FPN	
uh60m/cdu/CDU1_DAT	
uh60m/cdu/CDU1_INI	
uh60m/cdu/CDU1_XPDR	
uh60m/cdu/CDU1_ZRO	
uh60m/cdu/CDU1_ENT	
uh60m/cdu/CDU1_NXT	
uh60m/cdu/CDU1_PRV	
uh60m/cdu/CDU1_DIM	
uh60m/cdu/CDU1_BRT	
uh60m/cdu/CDU1_N0	
uh60m/cdu/CDU1_N1	
uh60m/cdu/CDU1_N2	
uh60m/cdu/CDU1_N3	
uh60m/cdu/CDU1_N4	
uh60m/cdu/CDU1_N5	
uh60m/cdu/CDU1_N6	
uh60m/cdu/CDU1_N7	
uh60m/cdu/CDU1_N8	
uh60m/cdu/CDU1_N9	
uh60m/cdu/CDU1_NMIN	
uh60m/cdu/CDU1_NDOT	
uh60m/cdu/CDU1_LFT	Cursor left
uh60m/cdu/CDU1_RGT	Cursor right
uh60m/cdu/CDU1_UP	Cursor up
uh60m/cdu/CDU1_DN	Cursor down

Mission commands

uh60m/mission/start	Start mission
uh60m/mission/stop	Stop/Hold mission
uh60m/mission/load	Load mission from file
uh60m/mission/download	Download mission from server
uh60m/mission/save	
uh60m/mission/reset	

uh60m/mission/clear	
uh60m/mission/add_wpt	Add current position to WPT list
uh60m/mission/skip	Skip current wpt or task
uh60m/mission/delete	
uh60m/mission/insert	
uh60m/mission/add	

Airfuel commands

uh60m/ops/c130_show	Toggle aerial refueling C130
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Datarefs

uh60m/cdu/mission_name	Name of selected mission file name
uh60m/door/preset	Selected door preset
uh60m/conf/probe	Refuel probe installed
uh60m/conf/probe_ext	Refuel probe extended
uh60m/conf/hoist	Hoist winch installed
uh60m/conf/hrst	Rope mount installed
uh60m/conf/doors	Cockpit doors installed
uh60m/conf/seats	Cargo seats installed
uh60m/conf/cargo	
uh60m/conf/guns	Guns installed
uh60m/conf/pilots	Pilots installed
uh60m/conf/flir	FLIR cam installed
uh60m/conf/earplugs	Earplug / Headset
uh60m/sys/sw/hstab	HSTAB switch
uh60m/sys/sw/sas1	SAS1 switch
uh60m/sys/sw/sas2	SAS2 switch
uh60m/sys/sw/sas_boost	SAS boost switch
uh60m/sys/sw/tailwheel	Tailwheel lock switch
uh60m/sys/sw/egi1	EGI 1 switch
uh60m/sys/sw/egi2	EGI 2 switch
uh60m/sys/sw/ralt_test	Radio altimeter test
uh60m/sys/sw/hydr_bkup	Hydraulic backup switch

uh60m/sys/sw/air_src	Air source for starter
uh60m/sys/sw/stby_inst	Backup / Standby instrument
uh60m/sys/sw/lamp_test	Lamp test
uh60m/sys/sw/fire_test	Fire test
uh60m/sys/hover_mode	HSI Hover mode
uh60m/sys/hover_hold	Hover hold state
uh60m/sys/level_hold	Level hold state
uh60m/sys/fps_on	FPS / Heading-hold
uh60m/cam/selected	Selected camera
uh60m/ops/winch_ena	Winch enabled
uh60m/weapons/flares_armed	CHAFF/FLARE armed
uh60m/mfd/on	Array/4 – MFD on
uh60m/mfd/page	Array/4 – MFD page
uh60m/mfd/tgt_mode	
uh60m/mfd/hsi_mode	
uh60m/mfd/show_efis	
uh60m/mfd/show_wxr	Show EFIS weather radar
uh60m/sys/CB	Array/100 – circuit breakers

functional CBs

CB_hstab = 8
 CB_sas = 9
 CB_vhf1 = 12 → com1 & com2 power
 CB_tail_lock = 13
 CB_master_wrn = 15
 CB_egi1 = 34 → gps power
 CB_egi2 = 16 → gps2 power
 CB_hydr_pump = 24